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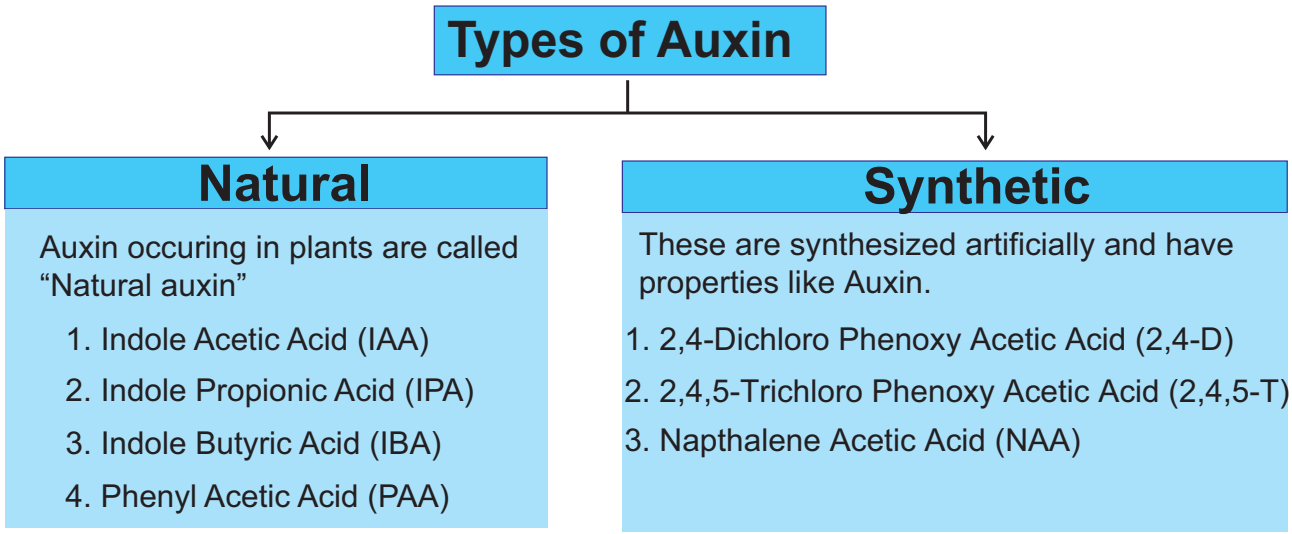


Figure 15.11: Classification of Auxins

2. Occurrence

Auxin is generally produced by the growing tips of the stem and root, from where they migrate to the region of the action.

3. Types of Auxin

Auxins are divided into two categories Natural auxins and Synthetic auxins (Figure 15.11).

Anti-auxins

Anti-auxin compounds when applied to the plant inhibit the effect of auxin. Example: 2, 4, 5-Tri Iodine Benzoic Acid (TIBA) and Napthylpthalamine.

4. Free auxin

They move out of tissues as they are easily diffusible. Example: IAA.

5. Bound Auxin

They are not diffusible. Example: IAA-Aspartic acid

6. Precursor

The amino acid Tryptophan is the precursor of IAA and zinc is required for its synthesis.

7. Chemical structure

Auxin has similar chemical structure of IAA.

8. Transport in Plants

Auxin is polar in transport. It includes basipetal and acropetal transport. Basipetal means transport through phloem from shoot to root and acropetal means transport through xylem from root to shoot.

9. Bioassay (Avena Curvature Test / Went Experiment)

Bioassay means testing of substances for their activity in causing a growth response in a living plant or its part.

The procedure involves the following steps:

When the *Avena* seedlings have attained a height of 15 to 30 mm, about 1mm of the coleoptile tip is removed. This apical part is the source of natural auxin. The tip is now placed on agar blocks for few hours. During this period, the auxin diffuses out of these tips into the agar. The auxin containing agar block is now